

**Name : Vaibhav Kore (TE COMP-3, R Batch) Roll No- 70(PRN F24123007) AI
Practical No – 5
Develop elementary chatbot for any suitable customer interaction application.**

Code :-

```
package ai;
import javax.swing.*;
import javax.swing.border.EmptyBorder;
import java.awt.*;
import java.awt.event.*;
import java.util.HashMap;
import java.time.LocalDateTime;
import java.time.format.DateTimeFormatter;

public class Chatbot extends JFrame implements ActionListener {

    JPanel chatPanel, sidePanel;
    JTextField inputField;
    JButton sendButton;
    JScrollPane scrollPane;
    JLabel userLabel;

    HashMap<String, String> responses = new HashMap<>();
    String userName = "";

    public Chatbot() {
        setTitle("Chatbot by Vaibhav");
        setSize(700, 550);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLayout(new BorderLayout());

        // Sidebar
        sidePanel = new JPanel();
        sidePanel.setPreferredSize(new Dimension(180, 0));
        sidePanel.setBackground(new Color(25, 25, 50));
        sidePanel.setLayout(new BoxLayout(sidePanel, BoxLayout.Y_AXIS));

        JLabel title = new JLabel("CHATBOT");
        title.setForeground(Color.WHITE);
        title.setFont(new Font("Arial", Font.BOLD, 20));
        title.setBorder(new EmptyBorder(20, 20, 10, 20));

        JLabel subtitle = new JLabel("by Vaibhav");
        subtitle.setForeground(Color.LIGHT_GRAY);
        subtitle.setBorder(new EmptyBorder(0, 20, 20, 20));

        JLabel userSection = new JLabel("User:");
        userSection.setForeground(new Color(0, 200, 255));
        userSection.setBorder(new EmptyBorder(10, 20, 5, 20));

        userLabel = new JLabel("Not logged in");
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userLabel.setForeground(Color.WHITE);
userLabel.setBorder(new EmptyBorder(0, 20, 10, 20));

JLabel featureTitle = new JLabel("Features:");
featureTitle.setForeground(new Color(0, 200, 255));
featureTitle.setBorder(new EmptyBorder(10, 20, 5, 20));

JLabel features = new JLabel("<html>- Chatting<br>- Date & Time<br>- GK Questions<br>- Cricket Info<br>- Programming Info</html>");
features.setForeground(Color.WHITE);
features.setBorder(new EmptyBorder(0, 20, 10, 20));

JLabel commandTitle = new JLabel("Commands:");
commandTitle.setForeground(new Color(0, 200, 255));
commandTitle.setBorder(new EmptyBorder(10, 20, 5, 20));

JLabel commands = new
JLabel("<html>hello<br>java<br>python<br>dhoni<br>time<br>date<br>bye</html>");
commands.setForeground(Color.WHITE);
commands.setBorder(new EmptyBorder(0, 20, 10, 20));

JLabel aboutTitle = new JLabel("About:");
aboutTitle.setForeground(new Color(0, 200, 255));
aboutTitle.setBorder(new EmptyBorder(10, 20, 5, 20));

JLabel about = new JLabel("<html>This is a GUI Chatbot<br>made using Java Swing.</html>");
about.setForeground(Color.WHITE);
about.setBorder(new EmptyBorder(0, 20, 20, 20));

sidePanel.add(title);
sidePanel.add(subtitle);
sidePanel.add(userSection);
sidePanel.add(userLabel);
sidePanel.add(featureTitle);
sidePanel.add(features);
sidePanel.add(commandTitle);
sidePanel.add(commands);
sidePanel.add(aboutTitle);
sidePanel.add(about);

// Chat Panel
chatPanel = new JPanel();
chatPanel.setLayout(new BoxLayout(chatPanel, BoxLayout.Y_AXIS));
chatPanel.setBackground(new Color(220, 230, 240));

scrollPane = new JScrollPane(chatPanel);
scrollPane.setBorder(null);

// Bottom input
inputField = new JTextField();
inputField.setFont(new Font("Arial", Font.PLAIN, 16));

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inputField.addActionListener(this);

sendButton = new JButton("Send");
sendButton.setBackground(new Color(0, 102, 204));
sendButton.setForeground(Color.WHITE);
sendButton.addActionListener(this);

JPanel bottomPanel = new JPanel(new BorderLayout());
bottomPanel.add(inputField, BorderLayout.CENTER);
bottomPanel.add(sendButton, BorderLayout.EAST);

add(sidePanel, BorderLayout.WEST);
add(scrollPane, BorderLayout.CENTER);
add(bottomPanel, BorderLayout.SOUTH);

loadResponses();
botMessage("Hello! What is your name?");

setVisible(true);
}

public void loadResponses() {
    responses.put("hello", "Hello!");
    responses.put("hi", "Hi!");
    responses.put("how are you", "I am fine. How can I help you?");
    responses.put("java", "Java is an object-oriented programming language.");
    responses.put("python", "Python is an easy programming language.");
    responses.put("ai", "AI stands for Artificial Intelligence.");
    responses.put("dbms", "DBMS is Database Management System.");
    responses.put("os", "Operating System manages computer hardware and software.");
    responses.put("dhoni", "MS Dhoni is the greatest captain of all time.");
    responses.put("ipl", "IPL is Indian Premier League.");
    responses.put("college", "College life is the best time to learn and grow.");
    responses.put("exam", "Study well and do your best. All the best!");
    responses.put("bye", "Goodbye!");
}

public String getTime() {
    return LocalDateTime.now().format(DateTimeFormatter.ofPattern("HH:mm:ss"));
}

public String getDate() {
    return LocalDateTime.now().format(DateTimeFormatter.ofPattern("dd-MM-yyyy"));
}

public void actionPerformed(ActionEvent e) {
    String text = inputField.getText();
    userMessage(text);

    if (userName.equals("")) {
        userName = text;
        userLabel.setText(userName);
    }
}

```

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        botMessage("Nice to meet you, " + userName + "!");
        inputField.setText("");
        return;
    }

    String response = getResponse(text.toLowerCase());
    botTyping(response);

    inputField.setText("");

    if (text.equalsIgnoreCase("bye")) {
        System.exit(0);
    }
}

public String getResponse(String input) {
    if (input.contains("time"))
        return "Current time is " + getTime();

    if (input.contains("date"))
        return "Today's date is " + getDate();

    for (String key : responses.keySet()) {
        if (input.contains(key))
            return responses.get(key);
    }

    return "I am still learning. Ask something else!";
}

public void userMessage(String text) {
    JPanel panel = new JPanel(new FlowLayout(FlowLayout.RIGHT, 5, 5));
    JLabel label = new JLabel("<html><p style='padding:6px>" + text + "</p></html>");
    label.setOpaque(true);
    label.setBackground(new Color(0, 102, 204));
    label.setForeground(Color.WHITE);
    label.setBorder(new EmptyBorder(5,10,5,10));
    panel.add(label);
    panel.setBackground(new Color(220, 230, 240));
    chatPanel.add(panel);
    scrollToBottom();
}

public void botMessage(String text) {
    JPanel panel = new JPanel(new FlowLayout(FlowLayout.LEFT, 5, 5));
    JLabel label = new JLabel("<html><p style='padding:6px>" + text + "</p></html>");
    label.setOpaque(true);
    label.setBackground(Color.WHITE);
    label.setBorder(new EmptyBorder(5,10,5,10));
    panel.add(label);
    panel.setBackground(new Color(220, 230, 240));
    chatPanel.add(panel);
}

```

```

        scrollToBottom();
    }

    public void botTyping(String response) {
        new Thread(() -> {
            botMessage("Typing...");
            try { Thread.sleep(700); } catch (Exception e) {}
            chatPanel.remove(chatPanel.getComponentCount() - 1);
            botMessage(response);
        }).start();
    }

    public void scrollToBottom() {
        chatPanel.revalidate();
        JScrollBar vertical = scrollPane.getVerticalScrollBar();
        vertical.setValue(vertical.getMaximum());
    }

    public static void main(String[] args) {
        new Chatbot();
    }
}

```

Output :

